



The *TOGAF*® Standard — *10th Edition*

**Information Architecture -
Business Intelligence &
Analytics**

The Open Group TOGAF® Series Guide

Table of Contents

Information Architecture - Business Intelligence & Analytics	iii
Preface	v
The Open Group	v
The TOGAF® Standard, a Standard of The Open Group	v
This Document	v
About the TOGAF® Series Guides	vi
Intended Audience	vi
Trademarks	vii
About the Authors	viii
Acknowledgements	ix
Referenced Documents	x
1. Introduction	1
1.1. The ArchiMate® Modeling Notation	1
2. Business Intelligence & Analytics Capability	3
2.1. BI & Analytics High-Level Reference Model	3
2.2. BI & Analytics High-Level Business Functions Description	3
2.3. BI & Analytics High-Level Application Functions Description	4
3. Process and Methodology	5
3.1. TOGAF ADM Specialization for BI & Analytics Capability	5
3.2. Reference BI & Analytics Roadmap Steps (ADM Phase F – Migration Planning)	7
3.2.1. Step 1	8
3.2.2. Step 2	10
3.2.3. Step 3	12
3.2.4. Step 4	15
4. Reference Models	18
4.1. BI & Analytics Reference Model – Goals	18
4.2. BI & Analytics Reference Model – Detailed Business Functions	21
4.2.1. Description of the Business Functions	22
4.3. BI & Analytics Reference Model – Detailed Application Functions	25
4.3.1. Description of the Application Functions	25
4.4. BI & Analytics Reference Model – Requirements	29
A: Glossary	31
Index	34

Information Architecture - Business Intelligence & Analytics

The Open Group TOGAF® Series Guide

Prepared by Céline Lescop and Jean-Baptiste Piccirillo, with the support of the
AXA Data Architecture Community and The Open Group Information Architecture Work Group

Evaluation Copy

Copyright © 2025, The Open Group. All rights reserved.

No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without the prior permission of the copyright owners. Specifically, without such written permission, the use or incorporation of this publication, in whole or in part, is NOT PERMITTED for the purposes of training or developing large language models (LLMs) or any other generative artificial intelligence systems, or otherwise for the purposes of using, or in connection with the use of, such technologies, tools, or models to generate any data or content and/or to synthesize or combine with any other data or content.

Any use of this publication for commercial purposes is subject to the terms of the Annual Commercial License relating to it. For further information, see www.opengroup.org/legal/licensing.

The Open Group TOGAF® Series Guide

Information Architecture - Business Intelligence & Analytics

ISBN: 1-957866-28-4

Document Number: G238

Published by The Open Group, August 2023.

Updated May 2025 to apply editorial changes as part of Technical Corrigendum 1 update.

Comments relating to the material contained in this document may be submitted to:

The Open Group, Apex Plaza, Forbury Road, Reading, Berkshire, RG1 1AX, United Kingdom
or by electronic mail to:

ogspeccs@opengroup.org

Built with [asciidocctor](#), version 2.0.18. Backend: pdf Build date: 2025-06-04 10:22:13 +0100

Preface

The Open Group

The Open Group is a global consortium that enables the achievement of business objectives through technology standards and open source initiatives by fostering a culture of collaboration, inclusivity, and mutual respect among our diverse group of 900+ memberships. Our membership includes customers, systems and solutions suppliers, tool vendors, integrators, academics, and consultants across multiple industries.

The mission of The Open Group is to drive the creation of Boundaryless Information Flow™ achieved by:

- Working with customers to capture, understand, and address current and emerging requirements, establish policies, and share best practices
- Working with suppliers, consortia, and standards bodies to develop consensus and facilitate interoperability, to evolve and integrate specifications and open source technologies
- Offering a comprehensive set of services to enhance the operational efficiency of consortia
- Developing and operating the industry's premier certification service and encouraging procurement of certified products

Further information on The Open Group is available at www.opengroup.org.

The Open Group publishes a wide range of technical documentation, most of which is focused on development of Standards and Guides, but which also includes white papers, technical studies, certification and testing documentation, and business titles. Full details are available at www.opengroup.org/library.

The TOGAF® Standard, a Standard of The Open Group

The TOGAF® Standard is a proven enterprise methodology and framework used by the world's leading organizations to improve business efficiency.

This Document

This document is the TOGAF® Series Guide to Information Architecture: Business Intelligence & Analytics. It has been developed and approved by The Open Group.

This document is structured as follows:

- **Chapter 1** provides a high-level introduction to this document, introducing the BI & Analytics topics
- Section describes the ArchiMate® modeling notation as used in this document
- **Chapter 2** describes the BI & Analytics capability, introducing the reference model from a high-level point of view
- **Chapter 3** proposes an Architecture Development Method (ADM) specialization for BI & Analytics and data platforms
- **Chapter 4** proposes detailed functional reference models describing BI & Analytics capabilities

About the TOGAF® Series Guides

The TOGAF® Series Guides contain guidance on how to use the TOGAF® Standard and how to adapt it to fulfill specific needs.

The TOGAF® Series Guides are expected to be the most rapidly developing part of the TOGAF® Standard and are positioned as the guidance part of the standard. While the TOGAF® Fundamental Content is expected to be long-lived and stable, guidance on the use of the TOGAF® Standard can be industry, architectural style, purpose, and problem-specific. For example, the stakeholders, concerns, views, and supporting models required to support the transformation of an extended enterprise may be significantly different than those used to support the transition of an in-house IT environment to the cloud; both will use the Architecture Development Method (ADM), start with an Architecture Vision, and develop a Target Architecture on the way to an Implementation and Migration Plan. The TOGAF® Fundamental Content remains the essential scaffolding across industry, domain, and style.

Intended Audience

The intended audience for this TOGAF® Series Guide is Enterprise Architects, Data Architects, and any IT professional working on data BI topics, etc.

More information is available, along with a number of tools, guides, and other resources, at www.opengroup.org/architecture.

Trademarks

ArchiMate, FACE, FACE logo, Future Airborne Capability Environment, Making Standards Work, Open Footprint, Open O logo, Open O and Check certification logo, Open Subsurface Data Universe, OSDU, SOSA, SOSA logo, The Open Group, TOGAF, UNIX, UNIXWARE, and X logo are registered trademarks and Boundaryless Information Flow, Build with Integrity Buy with Confidence, Commercial Aviation Reference Architecture, Dependability Through Assuredness, Digital Practitioner Body of Knowledge, DPBoK, EMMM, FHIM Profile Builder, FHIM logo, FPB, IT4IT, IT4IT logo, O-AA, O-DA, O-DEF, O-HERA, O-PAS, O-TTPS, O-VBA, Open Agile Architecture, Open FAIR, Open Process Automation, Open Trusted Technology Provider, Sensor Integration Simplified, and Sensor Open Systems Architecture are trademarks of The Open Group.

Hadoop is a registered trademark of the Apache Software Foundation (ASF).

Java is a registered trademark of Oracle and/or its affiliates.

All other brands, company, and product names are used for identification purposes only and may be trademarks that are the sole property of their respective owners.

Evaluation Copy

About the Authors

AXA Data Architecture Community

This document is inspired by Data Architecture deliverables donated to The Open Group by AXA. They are the results of the Data Architects community's work led by the group and with the contribution of affiliates mainly in France, Germany, Belgium, United Kingdom, Japan, US, Hong Kong, Spain, Switzerland, and Italy from 2015 to 2021. The aim of this community is to share Data Architecture best practices across the AXA organization to improve and accelerate the architecture delivery.

AXA is a global insurer (www.axa.com).

Céline Lescop – Author

Céline holds a Master's degree (1996) in Computer Science (www.ensiie.fr). She was a Data Project Leader in the pharmaceutical industry from 1999 until 2012 (Galderma & Astrazeneca), an Architect and TOGAF® trainer from 2012 to 2015 at Arismore (The Open Group, France), and has been an Enterprise Architect at AXA leading Data Architecture practice since 2016 and the Digital Sustainability program since 2020.

Jean-Baptiste Piccirillo – Author

Jean-Baptiste holds a Master's degree (2010) in Computer Science (www.ensiie.fr). He was an Architect until 2015 for Arismore (The Open Group, France), a Data Scientist for AXA from 2015 to 2018, and he is currently a Data Architecture and Data Management Consultant for Rhapsodies Conseil (www.rhapsodiesconseil.fr) supporting AXA.

Acknowledgements

(Please note affiliations were current at the time of approval.)

The Open Group gratefully acknowledges members of The Open Group Architecture Forum past and present for their contribution in the development of this document:

- Berenger Caudron, Data Architect at AXA Group Operation
- Chris Harding, Principal, Lacibus Ltd.
- Guillaume Hervouin, Data Architect at AXA Group Operation
- Jann-Philipp Zocher, Data Architect at AXA Germany

The Open Group gratefully acknowledges the following reviewers who participated in the Company Review of this document:

- Chris Harding
- James Thomas

Evaluation Copy

Referenced Documents

The following documents are referenced in this TOGAF® Series Guide:

- [C220] The TOGAF® Standard, 10th Edition, a standard of The Open Group (C220), published by The Open Group, April 2022; refer to: www.opengroup.org/library/c220
- [C226] ArchiMate® 3.2 Specification, a standard of The Open Group (C226), published by The Open Group, October 2022; refer to: www.opengroup.org/library/c226
- [G176] TOGAF® Series Guide: Business Scenarios (G176), published by The Open Group, April 2022; refer to: www.opengroup.org/library/g176
- [Inmon, 1995] What is a Data Warehouse?, by W. H. Inmon, Prism, Volume 1, Number 1, 1995

Evaluation Copy

1. Introduction

This document describes information architecture techniques to support architecture work using Business Intelligence (BI) & Analytics.

This document includes:

- Reference models for the assessment and design of BI & Analytics capabilities; see [Chapter 2](#), [Section 4.2](#), and [Section 4.3](#)
- An adaptation of the TOGAF ADM to support the BI & Analytics capability; see [Section 3.1](#) and [Section 3.2](#)

These best practices are based on project experiences and have been generalized to support wide usage in any organization. They aim to:

- Accelerate architecture delivery to make it more efficient by not reinventing the wheel
- Bring clarity on the topic, providing a framework to train Data Architects and other data professionals
- Provide a common language to describe BI & Analytics capabilities to data professionals: Chief Data Officers (CDOs), Data Managers, Architects, Data Engineers, Data Scientists, etc.

1.1. The ArchiMate® Modeling Notation

The reference models in this document are described using the ArchiMate modeling notation, as defined in the ArchiMate Specification, a standard of The Open Group [[C226](#)].

The ArchiMate elements used in this document are shown in [Figure 1-1](#).

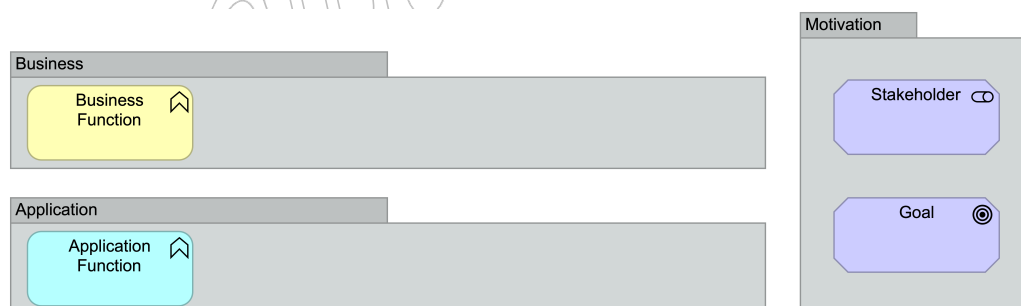


Figure 1-1: The ArchiMate Modeling Notation Elements Used in this Document

The ArchiMate definitions of each concept used in this document are as follows:

Motivation Aspect

- **Goal:** A high-level statement of intent, direction, or desired end state for an organization and its stakeholders.

In principle, a goal can represent anything a stakeholder may desire, such as a state of affairs, or a

produced value.

- **Stakeholder:** A stakeholder represents the role of an individual, team, or organization (or classes thereof) that represents their interests in the effects of the architecture.

A stakeholder has one or more interests in, or concerns about, the organization and its Enterprise Architecture. In order to direct efforts to these interests and concerns, stakeholders change, set, and emphasize goals. Stakeholders may also influence each other.

Business Layer

- **Business Function:** A business function represents a collection of business behavior based on a chosen set of criteria (typically required business resources and/or competencies), closely aligned to an organization, but not necessarily explicitly governed by the organization

Application Layer

- **Application Function:** An application function represents automated behavior that can be performed by an application component

Evaluation Copy

2. Business Intelligence & Analytics Capability

The BI & Analytics capability refers to the people, processes, organizations, and systems put together to create value-added information from relevant decision-making data of the organization. This capability breaks applicative data siloes and allows the structuring and leveraging of data according to decision-making and analytics use cases. It includes business value, data quality, availability, and security to deliver the required common data to relevant reporting or analytics use cases.

This capability is founded on reference business functions and reference application functions. This chapter covers the high-level functions reference model, and their definitions. More detailed functions are available in the dedicated reference model part of this document; see [Section 4.2](#) and [Section 4.3](#).

2.1. BI & Analytics High-Level Reference Model

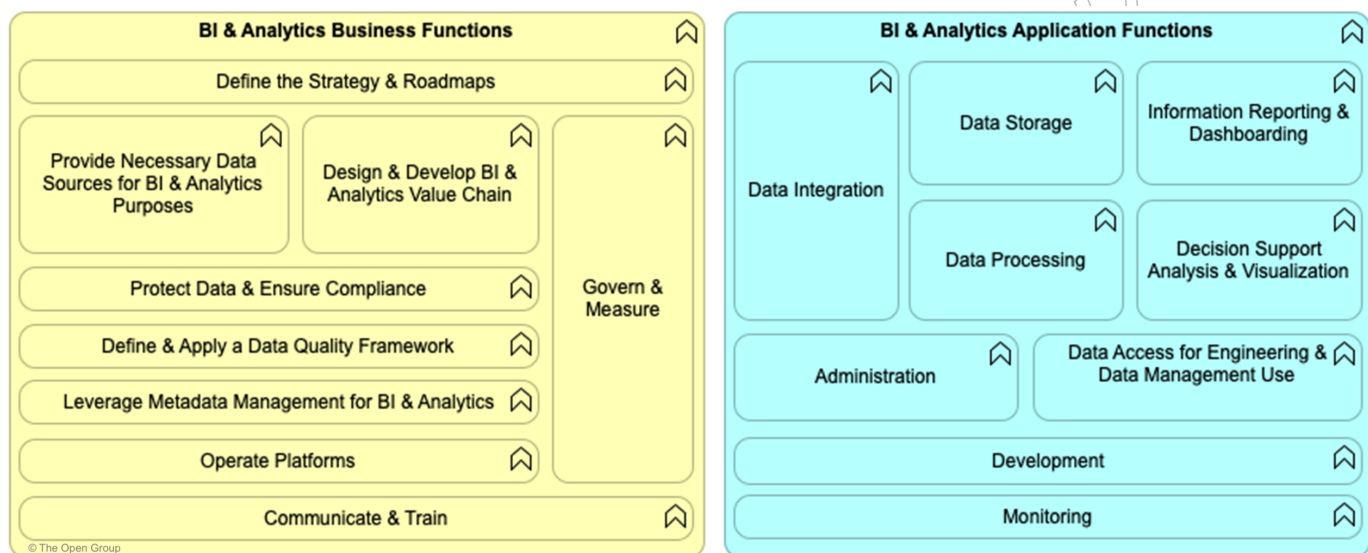


Figure 2-1: BI & Analytics Capability – High-Level Functions

2.2. BI & Analytics High-Level Business Functions Description

Table 2-1: BI & Analytics High-Level Business Functions

Business Function	Description
Define the Strategy & Roadmaps	Define the BI & Analytics vision for the organization, and how it contributes to organization objectives. Concretize this vision through a well-defined BI & Analytics strategy and roadmap.
Provide Necessary Data Sources for BI & Analytics Purposes	Identify the necessary data sources needed by the BI & Analytics strategy. Make the data available for related BI & Analytics use cases.
Design & Develop BI & Analytics Value Chain	Design and implement the data value chain to transform raw data into a “ready-to-use” business view serving BI & Analytics use cases.
Protect Data & Ensure Compliance	Put in place all processes and tools to protect available data and ensure compliance; e.g. General Data Protection Regulation (GDPR).

Business Function	Description
Define & Apply a Data Quality Framework	Put in place all processes and tools to ensure that the quality of provided data is at the expected level regarding business requirements.
Leverage Metadata Management for BI & Analytics	Implement processes and tools to leverage metadata management capabilities to document data and data lineage on the BI & Analytics value chain from operational data sources to data consumption.
Operate Platforms	Operate technical functions supporting BI & Analytics capability.
Govern & Measure	Govern the transformation of the BI & Analytics capability: measure data volumes, costs, carbon footprint, and actual value.
Communicate & Train	Communicate around BI & Analytics value, organize training plan for technical community (Data Engineer, Data Scientist, ...) and business end-users.

2.3. BI & Analytics High-Level Application Functions Description

Table 2-2: BI & Analytics High-Level Application Functions

Application Function	Description
Data Integration	Services that allow the integration of different data from one or more providers to enable data ingestion by consumers and data distribution for BI & Analytics purposes. This could be done, for example, by batch, streaming, interface, etc.
Data Storage	Services that allow the persistent storage of structured, semi-structured, or unstructured data.
Data Processing	Services that allow data transformation, including basic preparations or more advanced features, calculations, leveraging statistics, or data science.
Information Reporting & Dashboarding	Services allowing reporting and dashboarding activities using data stored in BI & Analytics.
Decision Support Analysis & Visualization	Services allowing data discovery and visualization using data stored in BI & Analytics.
Administration	Administration tasks required to support BI & Analytics capabilities, such as access management, scheduling, etc.
Data Access for Engineering & Data Management Use	Direct access to the data of BI & Analytics for advanced data stewards, data engineers, data scientists, data administrators, etc.
Development	Development capabilities including Integrated Development Environments (IDE) development integrated environment and continuous delivery chains for data scientists, data engineers, etc.
Monitoring	Monitoring services for infrastructure, network, and applications to guarantee the agreed service level.

3. Process and Methodology

3.1. TOGAF ADM Specialization for BI & Analytics Capability

A synthesis of key actions related to the TOGAF ADM phases (Preliminary, A, B, C, D, E, F, G) that should be done in the context of a BI & Analytics capability delivery is described in this chapter.

Table 3-1: BI & Analytics Capability Delivery Related to TOGAF ADM Phases

TOGAF ADM Phase	BI & Analytics Recommendations
Preliminary	Confirm that the proper Enterprise Architecture capabilities are in place to implement the model: repository, data sources, platforms, etc. Check that a multi-disciplinary team is in place with the prerequisite skills to architect an effective BI & Analytics capability.
A – Architecture Vision	Identify the relevant principles for BI & Analytics architecture and define them. Develop an overview of existing pain points related to reporting, data analytics, data science, and statistics data from a BI & Analytics user point of view (the TOGAF Business Scenario technique, can be used for that purpose [G176]). Develop the value proposition for the target BI & Analytics capability considering the existing or new business usages, improvement expected (data available, data quality, data documentation, data accessibility).
B – Business Architecture	Identify the as-is and to-be business users of the BI & Analytics capability. List the BI & Analytics usages on this business scope: futures, or existing usages to be improved. Define the target operational business organization to manage a BI & Analytics usage, from design to continuous deployment and maintenance. Define the value of each usage and the priority of each one, based, for example, on complexity and value for the organization. Identify the key data – at business object level first – required to deliver the expected use cases. Evaluate the coverage of the as-is capability and identify the gap with to-be requirements. The business functions provided in Section 2.2 can be used as a guide to the steps.

TOGAF ADM Phase	BI & Analytics Recommendations
C – Information Systems Architectures	Data: - Specify a data modeling method and its application to the enterprise data - From the data expected by the business, define the main attributes of each data structure and the links between them - Define the logical data structures that will be used to cover different business requirements – raw data structure, intermediate data layer (for example, enterprise/domain data warehouses), end data layer (data mart/denormalized business views) - For each set of data, define the lifecycle management (from integration in the platform to “end of life/deletion”) - Perform a data quality assessment of each set of data Application: - Define the Target Application Architecture to support collection, transport, storage, and access for BI & Analytics purposes - Identify the source applications providing the data expected by the business - Define the processes to support the growth of the platform in terms of data (volumes, diversity, duplicates) - Identify the gaps between as-is architecture and to-be in term of variety, velocity, volume The application functions provided in Section 2.3 can be used as a guide to the steps.
D – Technology Architecture	Identify requirements for data platforms to support BI & Analytics operations. Specify technology support for data modeling.
E – Opportunities and Solutions	Use the application reference models in this document to challenge solution providers around BI & Analytics. Reference models can be used as a basis to build selection criteria of the solutions.
F – Migration Planning	Refer to Section 3.2 to address the migration from a legacy BI & Analytics technical solution to a new generation one.
G – Implementation Governance	Organize compliance review with BI & Analytics projects to check that they respect BI & Analytics principles and the Target Architecture.

3.2. Reference BI & Analytics Roadmap Steps (ADM Phase F – Migration Planning)

Looking back at the history of BI & Analytics on the market, different architecture pattern styles and technical solutions have been used. For a given organization, the coexistence of several BI & Analytics capabilities using different technologies and duplicating data comes with difficulties if the following principles quoted in the TOGAF Standard [C220] are not respected:

- Principle 11: Data is Shared

Users have access to the data necessary to perform their duties; therefore, data is shared across enterprise functions and organizations (...) It is less costly to maintain timely, accurate data in a single application, and then share it, than it is to maintain duplicative data in multiple applications.

For a given Organization Information System, having the data kept first in an “Infocenter” and then copied in a “data warehouse” for other analytics is costly and complex.

- Principle 20: Control Technical Diversity

Technological diversity is controlled to minimize the non-trivial cost of maintaining expertise in and connectivity between multiple processing environments.

For a given Organization Information System maintaining both data warehouse running on Relational Database Management System (RDBMS) and Data Lake on Hadoop® comes with IT cost and complexity that should be justified with business value.

This section describes an approach to ensure that when a new technology/pattern is introduced for the BI & Analytics, the replacement of the legacy architecture by the new one is performed. Replacing a capability that is paid and up-and-running in an organization is never straightforward as it requires investment that is not easy to justify from a business point of view. This approach is agnostic of the technical solutions.

The following example deals with the shift from an enterprise data warehouse running on a relational database to new generation BI & Analytics data platforms that deliver the possibility to support volume x variety x velocity needs both for data storage (structured, unstructured, evens, etc.) and for data processing (file processing, SQL, graph processing, streams processing, machine learning, etc.).

Transition Architecture : A formal description of one state of architecture at an architecturally significant point in time.

Note: One or more Transition Architectures may be used to describe the progression in time from the Baseline to the Target Architecture [C220]. A Transition Architecture is defined as a relatively stable state of the architecture that exists for a limited period of time. Operations can run normally but the Transition Architecture does not realize the full vision that is covered by the Target Architecture.

Four Transition Architectures are described in the four following steps.

3.2.1. Step 1

Life before the new generation BI & Analytics data platform – a generic “as-is” situation is described here, but the level of maturity can vary a lot depending on the organization including no existing as-is enterprise data warehouse.

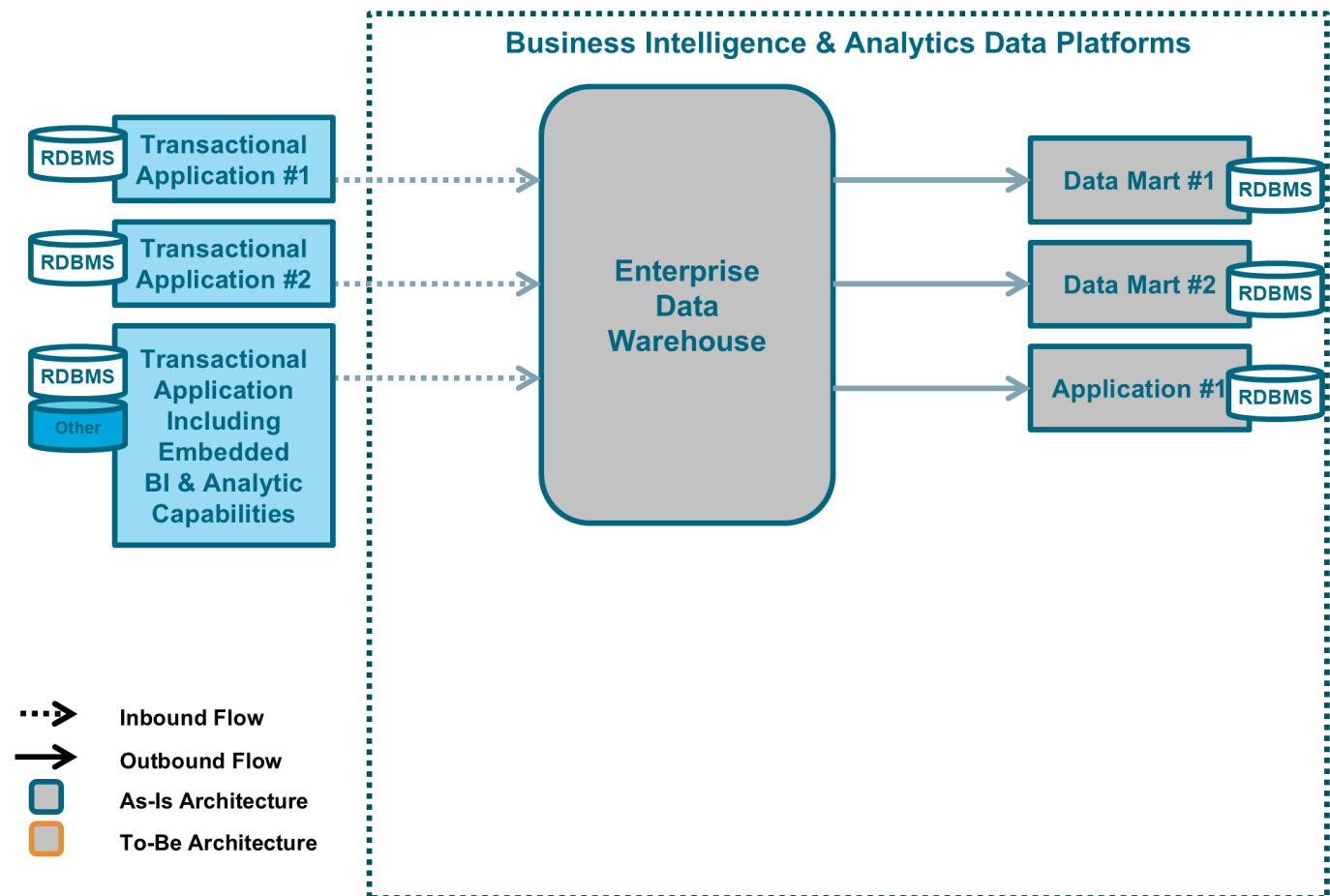


Figure 3-1: Step 1 – As-Is Architecture

- Transactional systems (for example, CRM, ERP, ...) stand alone with their databases
- Some transactional systems may include embedded BI & Analytics capabilities
- Some transactional systems based on an RDBMS contribute data to a data warehouse
- The data warehouse is the single point of truth and is the reference database of the enterprise historical data

In some organizations, the single point of storage may not be achieved in the as-is, and the full scope of data warehouse data is covered by several systems that are more or less interconnected.

Table 3-2: BI & Analytics Roadmap Step 1 Details

Benefits	One or more reference databases for RDBMS historical data and a single point of truth for BI
Missing Capabilities	Store raw data Analyze non-relational data Store analytical results datasets Process analytics “in-database” with no impact on the quality of service Shadow BI & Analytics to gather data that is not available in the data warehouse, at least for exploration and prototyping
Possible Triggers to Proceed to Step 2	Technical and/or cost limits on the volume of data processed by the RDBMS solutions Technical and/or cost limits on the volume of data stored by the RDBMS solutions Integration of new data in the data warehouse is too long and too expensive New analytics business needs (variety, volume, velocity) New requirement for an event processing application to provide recommendations based on the combination of events and historical data, like “next best customer action”, “fraud detection” ... Reduce shadow BI & Analytics

3.2.2. Step 2

[Transition step] The introduction of the new generation BI & Analytics data platforms.

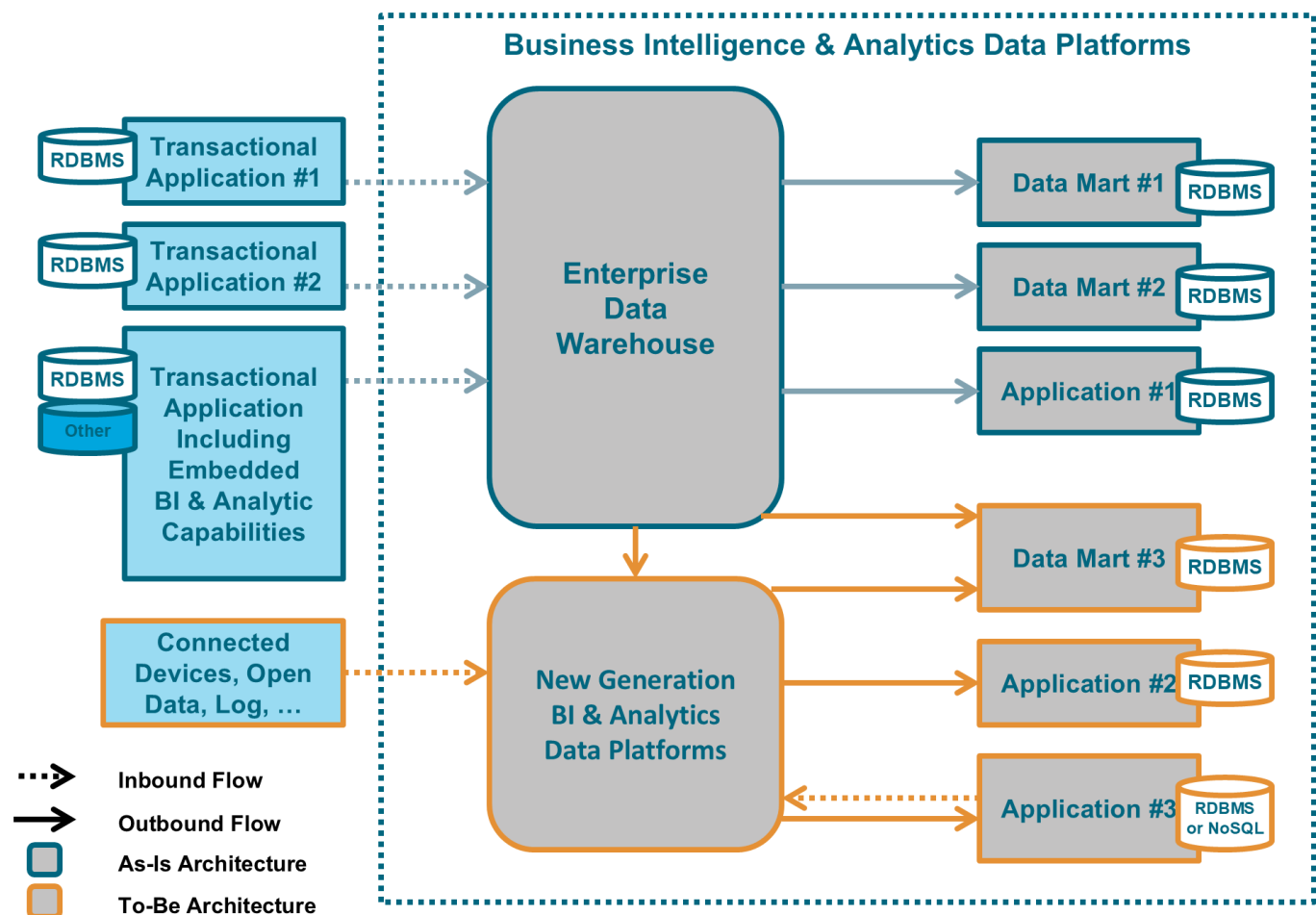


Figure 3-2: Step 2 – Introduction of the New Generation Platform

- The data warehouse feeds the new generation BI & Analytics data platforms with cleansed data, which they would not receive otherwise
- The new generation BI & Analytics data platform is used to address specific use cases that process new data sources (e.g., applications log files, connected device)
- In that Transition Architecture, some data marts, like Data Mart #3 in Figure 3-2, that have to be delivered in short delays to the business, are fed some data from the enterprise data warehouse, and also from the new generation BI & Analytics data platforms
- New BI & Analytics use cases can use new generation BI & Analytics data platform historical data and knowledge capabilities to improve decision-making during a business process

Those applications may create new data either through a human interface or by applying an algorithm on data. This new data might integrate the new generation BI & Analytics data platforms through the ingestion process to keep the history.

Table 3-3: BI & Analytics Roadmap Step 2 Details

Benefits	Analyze relational and non-relational data Access to a Data Sand Box New requirements for applications processing events to provide recommendations based on a combination of events and historical data like “next best customer action”, “fraud detection” ...
Missing Capabilities	Double storage of historical data: no single point of truth for analytics New generation BI & Analytics data platforms do not store raw data coming in from the data warehouse Data lineage is difficult due to the loss of raw data
Possible Triggers to Proceed to Step 3	Need for a single point of truth for analytics: requirement to cross all data Technical roadmap requires technical upgrade on existing solution that can be a good opportunity to migrate to new generation BI & Analytics data platforms Agility to meet new analytic requirement or application Requirement for data lineage Reduce shadow BI & Analytics

3.2.3. Step 3

[Transition step] The new generation BI & Analytics data platforms feed the data warehouse with new data, and cover the full scope of raw data.

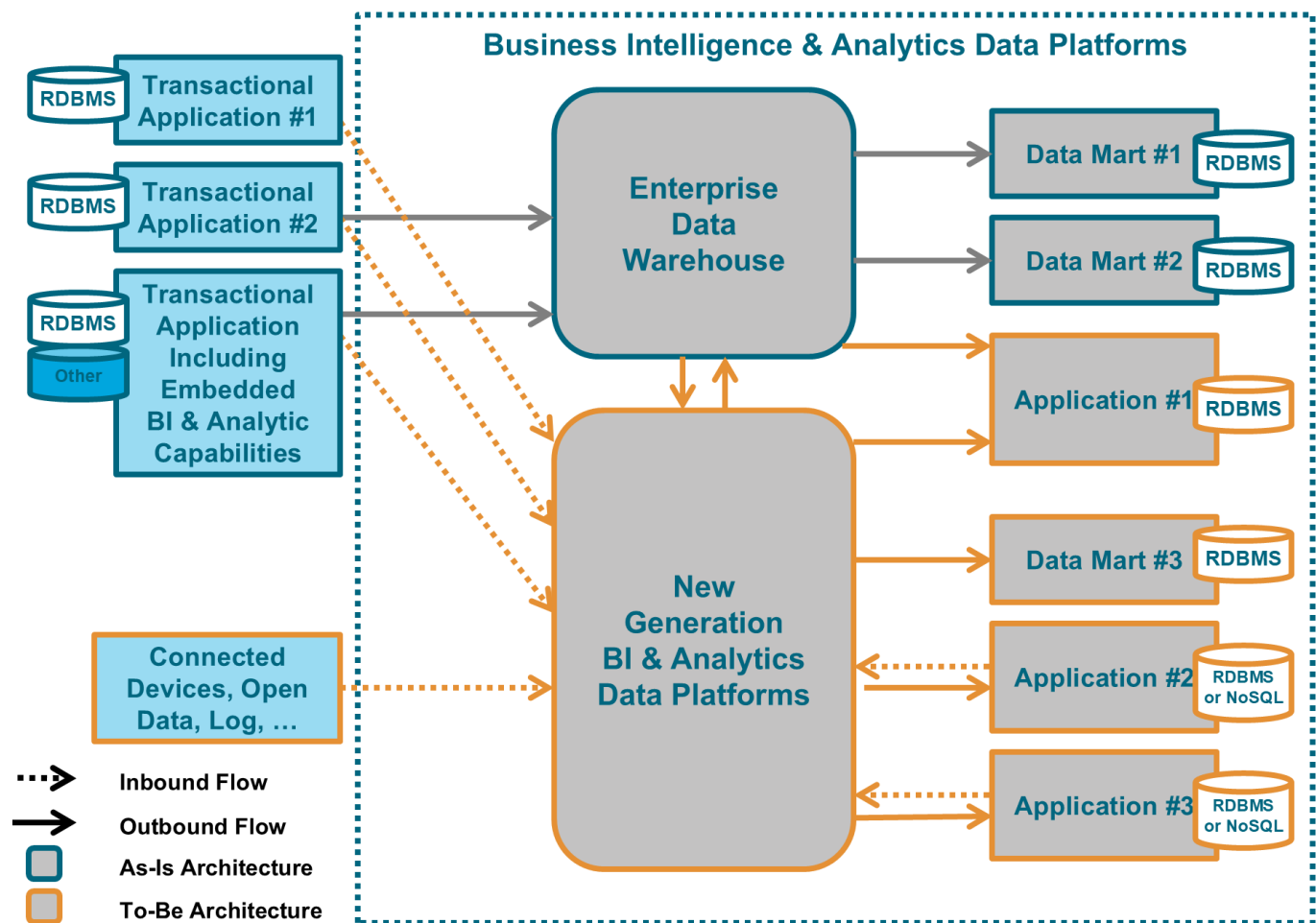


Figure 3-3: Step 3 – The New Generation Platform Grows

- Concerning data historically imported into the data warehouse, and also to the new generation BI & Analytics data platforms, the preferred approach is not to feed the data warehouse with the new generation BI & Analytics data platforms but for the data mart to go directly to Step 4 (nevertheless, this approach might be justified in some cases)
- In that step, data marts and applications are unplugged from the enterprise data warehouse to new generation BI & Analytics data platforms in an opportunistic and/or step-by-step approach, like “Data Mart #3” in [Figure 3-2](#)

In this transition step, some may now be fed by both the new generation BI & Analytics data platform and the enterprise data warehouse, like “Transactional Application #1” in [Figure 3-3](#).

- Data flows from transactional applications to the new generation BI & Analytics data platforms are different from those feeding the enterprise data warehouse, as they extract:
 - Raw data from the transactional application

- All the metadata from the transactional application including foreign keys, data dictionary

Such information is rarely explicit in the flow feeding the data warehouse. Indeed, that metadata may not be required in the data warehouse as some of this information is kept in the data warehouse modeling.

- All available data on a given business object, like a customer or a contract

This is not limited to the scope of data required by the analytic use case. This approach is a default one on the new generation BI & Analytics data platforms, and is allowed because of the low data cost storage.

- The new generation BI & Analytics data platforms act as the persistent raw data historical storage and information and knowledge preparation for the data warehouse replacing staging area or Operational Data Store (ODS)
- The opportunity of this Transition Architecture has to be checked against going directly to Step 4

Evaluation Copy

Table 3-4: BI & Analytics Roadmap Step 3 Details

Benefits from Previous Steps	Analyze relational and non-relational data Access to a Data Sand Box New requirement for an applications processing event to provide recommendations based on the combination of events and historical data, like “next best customer action”, “fraud detection”...
New Benefits	The new generation BI & Analytics data platforms store raw data, integrated data, and transformed/calculated data: all data marts can be plugged into the new generation BI & Analytics data platforms Data is available for data scientists in the Data Sand Box, increasing predictive capability and reducing shadow BI & Analytics Any new application requiring historical data can be built on the new generation BI & Analytics data platforms: agility to meet new analytic requirements Data lineage facilitated thanks to the storage of raw data
Missing Capabilities	Double storage of historical data: no single point of truth for analytics and double cost
Possible Triggers to Proceed to Step 4	Cost avoidance, optimization (storage, projects: work on data before modeling, ...) Double ingestion flow to the data warehouse and the new generation BI & Analytics data platforms to maintain Difficulties caused by the double storage of historical data: no single point of truth Opportunistic approach: technical roadmap or new business requests a major upgrade on the data warehouse architecture

3.2.4. Step 4

[Target Architecture] The new generation BI & Analytics data platforms feed all the data marts and cover all the BI & Analytics requirements from business users.

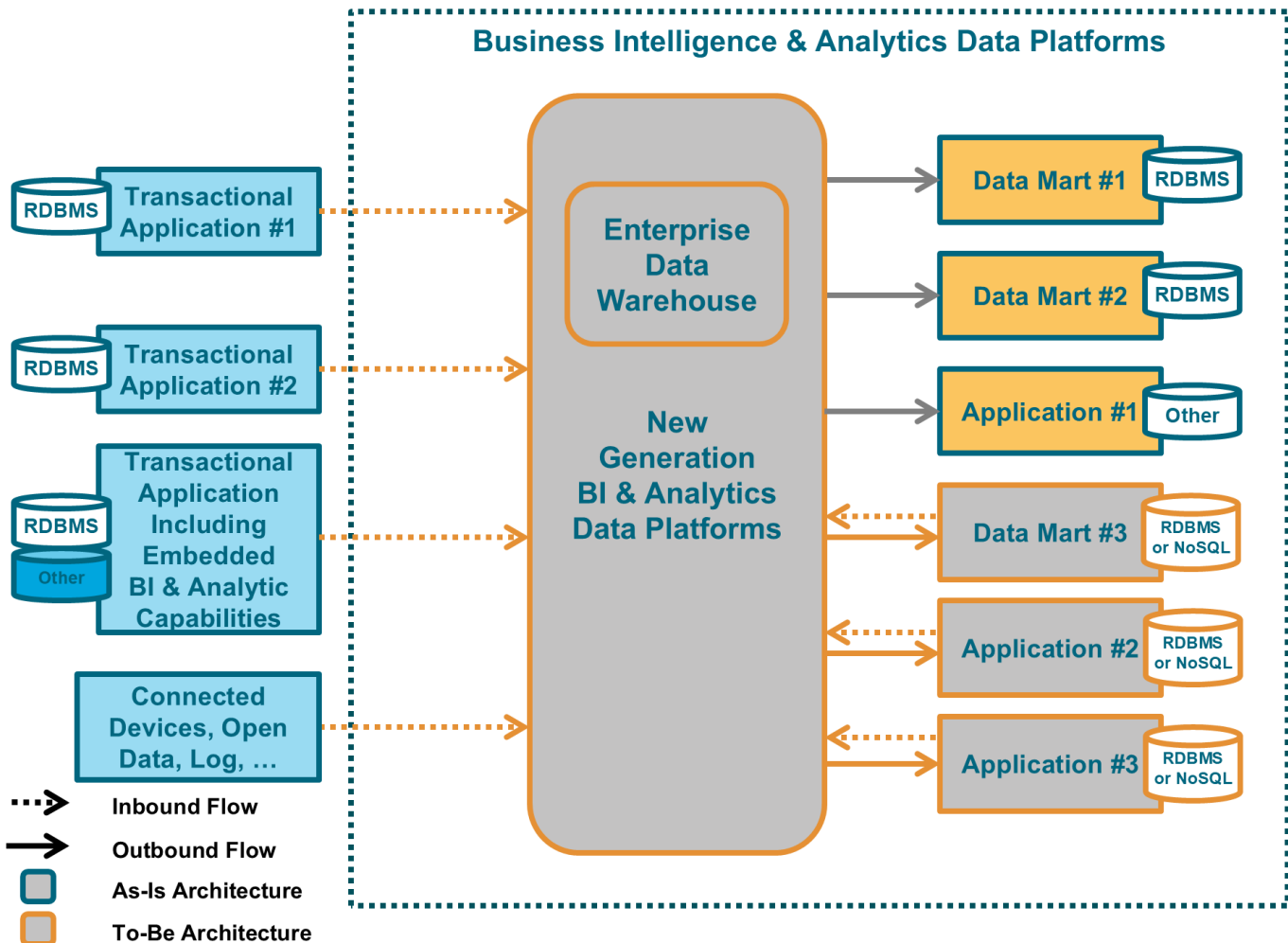


Figure 3-4: Step 4 – Target Architecture

Structurally, because of its capacity to store more volume, variety, and velocity, the new generation BI & Analytics data platforms can cover the scope of the data warehouse data and more. The conviction was established that having two repositories, data warehouse and new generation BI & Analytics data platforms for historical data, is not a reliable target vision. It will take time, but at the target new generation BI & Analytics data platforms will cover the full scope of the data warehouse, including the harmonized enterprise data model built to answer analytics requirements.

Existing data marts are plugged into the new generation BI & Analytics data platforms, which become the single point of truth for historical data storage.

Table 3-5: BI & Analytics Roadmap Step 4 Details

Benefits from the Previous Steps	Analyze relational and non-relational data Access to a Data Sand Box New requirement for an applications processing event to provide recommendations based on the combination of events and historical data, like “next best customer action”, “fraud detection” ... The new generation BI & Analytics data platforms store raw data, integrated data, and transformed/calculated data Data is available for data scientists in the Data Sand Box, increasing predictive capability and reducing shadow BI & Analytics Any new application requiring historical data can be built on the new generation BI & Analytics data platforms Data lineage facilitated thanks to the storage of raw data
New Benefits	No need to support maintenance on previous data warehouse infrastructure New generation BI & Analytics data platforms store raw data and all relevant data New generation BI & Analytics data platform is the single point of truth for historical data: reference database for historical raw data storage from which information and knowledge can be reprocessed any time, for any purpose All data stored in a single place: in principle we should not have a situation where some relevant data is not in the new generation BI & Analytics data platforms Investments in existing data marts and associated tools are preserved
Risks to Take into Account	There is a risk that some requirements met by the data warehouse architecture are not covered anymore in this step: to make that major shift possible, requirements currently covered in data warehouse have to be identified carefully and covered if required by the architecture based on the new generation BI & Analytics data platforms Data governance and metadata management will have to be piloted carefully in the new generation BI & Analytics data platforms to be able to leverage the full value of having all relevant raw data and information stored and secured in a single place

Step 4 is the proposed final target for a BI & Analytics capability. Steps 2 and 3 are possible Transition Architectures, but each organization will have to define its own Transition Architecture(s) when designing the level of its roadmap.

Indeed, the journey from Step 1 to Step 4 might be quite different depending on the current state of the organization:

- When the as-is architecture already includes an enterprise data warehouse, the shift from the data warehouse as the single point of truth for historical data storage might be longer than in an organization that has no enterprise data warehouse, and that will directly build the new generation BI & Analytics Target Architecture
- Not all organizations will be able to complete the journey within three years, but within five years the new generation BI & Analytics architecture should have taken a significant place in the architecture as the single point of truth for historical data storage

Information

Existing systems that result from big investments and that meet business requirements with an acceptable level of service will not be replaced until a technical migration is required or a new business need emerges. The objective is to focus first on added value and to maximize the use of past investments.

4. Reference Models

4.1. BI & Analytics Reference Model – Goals

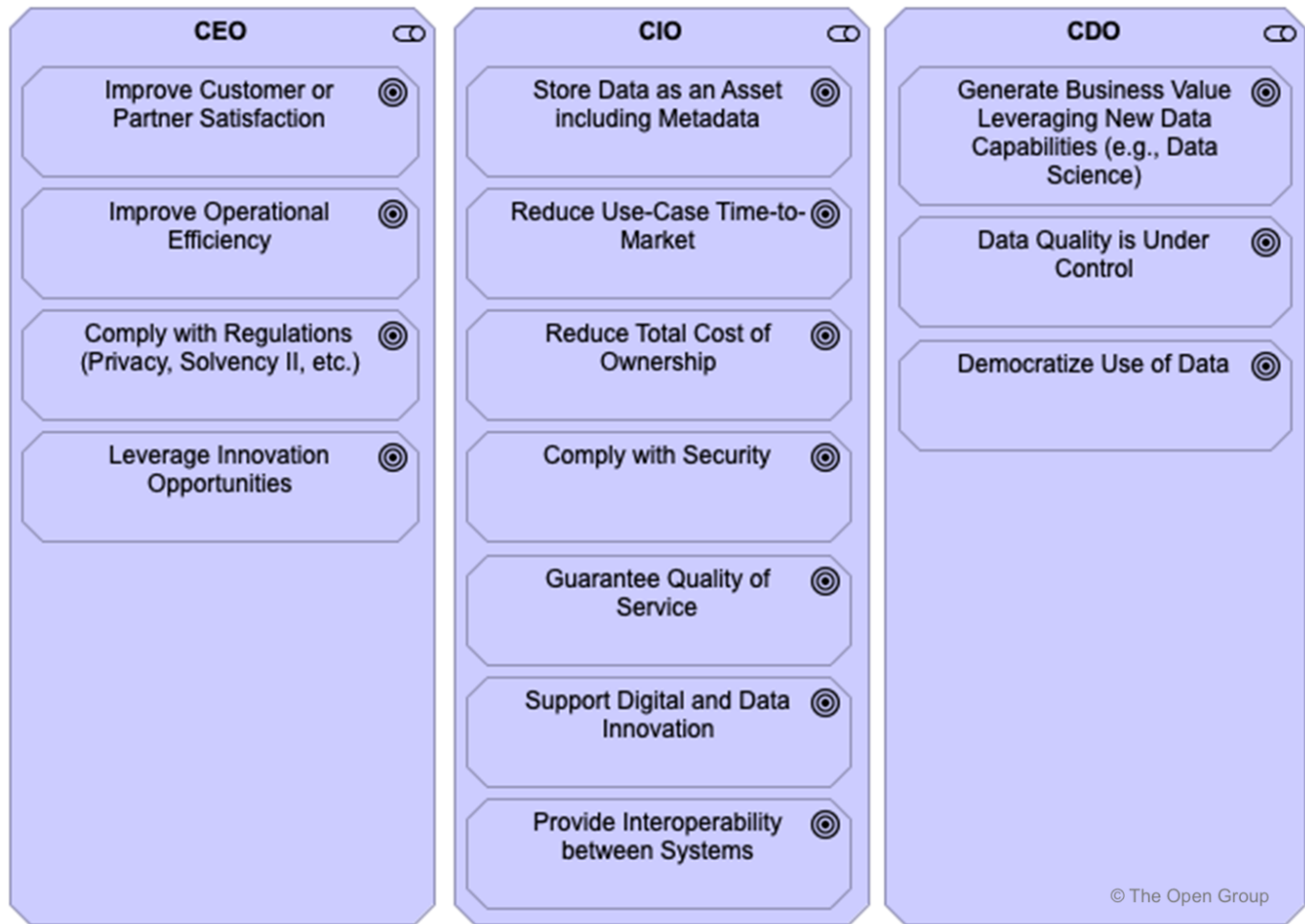


Figure 4-1: Goals Targeted by Stakeholders with BI & Analytics Capability

Table 4-1: Objectives and Goals of the BI & Analytics Reference Models

ID	Goals	Description	Key Performance Indicators (KPIs)
1	Comply with Regulations (e.g., Privacy, Solvency II, etc.)	Some regulations impact data management: - Data privacy regulation aims to guarantee customers' data ownership – all personal data needs to be sufficiently controlled to ensure its necessity, protection, restitution, and deletion (audits should be runnable on the system); in Europe this refers to European GDPR - Solvency II concerns the amount of capital that European Union insurance companies must hold to reduce the risk of insolvency	↑ number of datasets tagged with the level of privacy ↓ data leakages

ID	Goals	Description	Key Performance Indicators (KPIs)
2	Comply with Security – data classification and encryption	Data is classified as secret, confidential, public, etc. depending on its sensitivity and the evaluation of associated risk. It is then encrypted with encryption keys that are managed depending on this classification.	↓ security breaches
3	Comply with Security – data leakage	Data should be protected from eavesdropping, leakage, or theft according to its classification.	↓ security breaches
4	Comply with Security – identity and access management	Management of electronic or digital identities, including the organizational policies for managing digital identity as well as the technologies needed to support identity management.	↓ security breaches
5	Comply with Security – infrastructure baseline	A minimal security baseline to be provided for the infrastructure (administrator password changes on material delivered, port management, etc.).	↓ security breaches
6	Data Quality is Under Control	Data quality should remain at a suitable level to ensure the operational performance of the organization.	↑ number of datasets monitored by business quality rule
7	Democratize Use of Data	Users should have access to shared semantics and data required to perform their duties through tools appropriate to their analytics requirements – from ready-to-read reports to advanced queries or analytics. This principle of data democratization will continually “bump up against” the principle of data security. Under no circumstances will the data sharing principle cause confidential data to be compromised.	↑ number of decisions taken under data inputs
8	Guarantee Quality of Service	Includes availability; data should be available when needed and as appropriate.	↑ improve system time availability
9	Improve Customer or Partner Satisfaction	Use-cases and services that aim to increase customer or partner satisfaction.	↑ improve Net Promoter Score (NPS)
10	Improve Operational Efficiency	Use-cases and services that aim to increase agility in the organization, from innovation to operations.	↑ automated functions
11	Leverage Innovation Opportunities	The IT market is innovative. It provides opportunities to create new business models, generate revenues, or optimize.	↑ innovative features
12	Reduce Total Cost of Ownership	Increase cost efficiency.	↓ cost per user
13	Reduce Use-Case Time-to-Market	Setting up new use cases in an Agile manner to provide competitive advantage. This includes self-service for different stakeholders.	↓ delay from prototype validation to production running

ID	Goals	Description	Key Performance Indicators (KPIs)
14	Store Data as an Asset Including Metadata	Valuable data owned by the organization should be kept in safe storage and with appropriate documentation, and be available, understandable, and usable (qualitative, meaningful, query-able, process-able, etc.) for transversal analytics.	↑ number of permanent connections
15	Generate Business Value Leveraging New Data Capabilities (e.g., Data Science)	New data capabilities come with the promise of creating new business value and new markets. Organizations have to be able to compete in the international market leveraging new data capabilities. CDO positions have been created to make it happen.	↑ value generated by data use cases
16	Provide Interoperability Between Systems	Allow the break of organizational silos by ensuring application interoperability.	↑ number of APIs available and used
17	Support Digital and Data Innovation	Provide the digital capabilities to support innovation.	↑ innovation deployed at scale with Profit and Loss (P&L) impact

Evaluation Copy

4.2. BI & Analytics Reference Model – Detailed Business Functions

Figure 4-2 provides detailed business functions of the BI & Analytics capability.

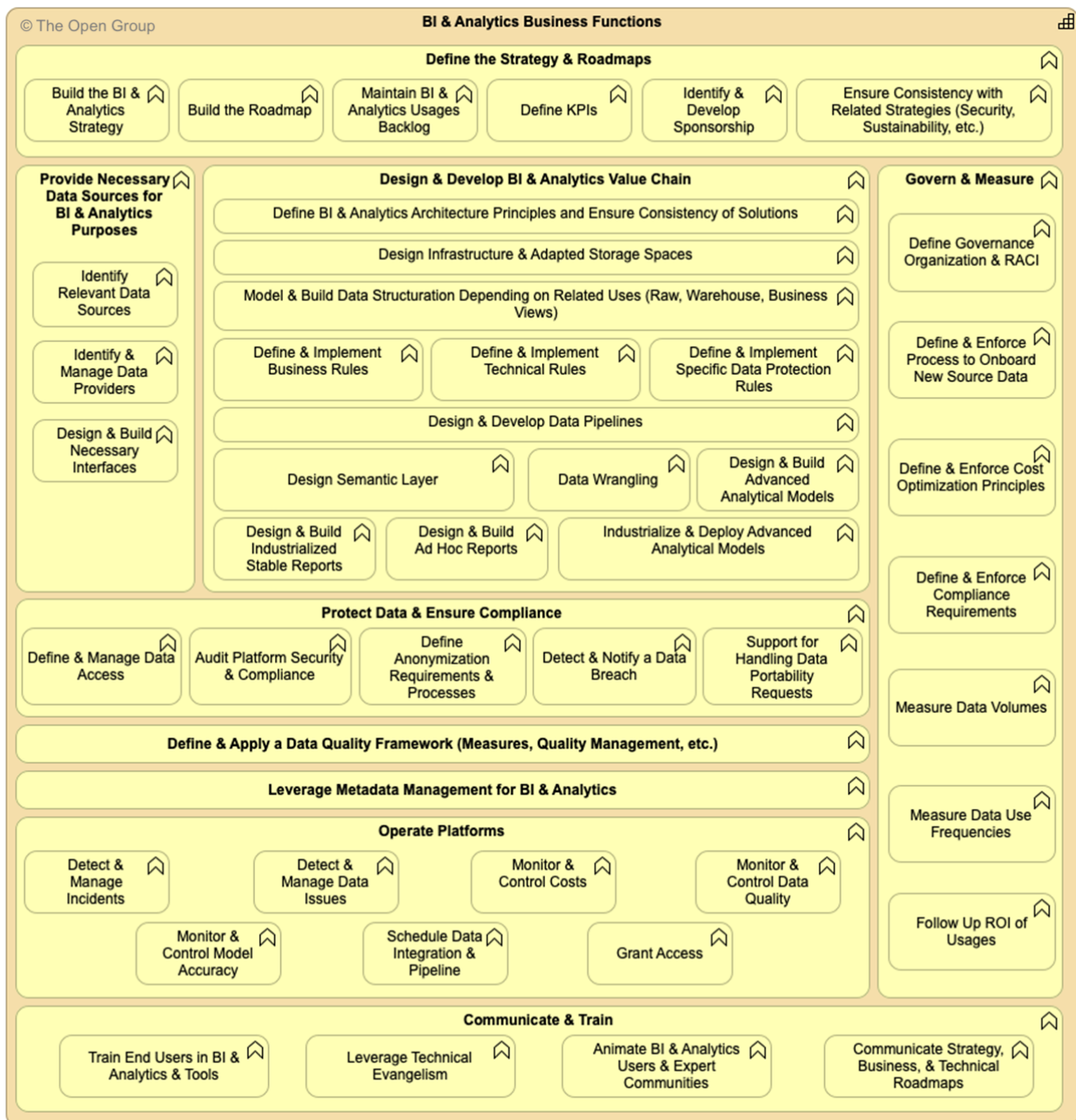


Figure 4-2: BI & Analytics Detailed Business Functions

4.2.1. Description of the Business Functions

In [Table 4-2](#), **bold lines are macro functions** and the others are related sub-functions.

Table 4-2: Detailed Business Functions

Business Function	Description
Define the Strategy & Roadmaps	
Build the BI & Analytics Strategy	Build business and IT strategy around BI & Analytics, having a defined target and overall journey.
Build the Roadmap	Build the roadmap to implement the BI & Analytics strategy.
Maintain BI & Analytics Usages Backlog	Maintain a list of qualified business usages that are and/or should be supported by BI & Analytics capabilities.
Define KPIs	Define KPIs showing how BI & Analytics reaches its strategic objectives (qualitative/quantitative).
Identify & Develop Sponsorship	Having business and IT sponsors implicated on BI & Analytics governance.
Ensure Consistency with Related Strategies (Security, Sustainability, etc.)	Ensure that BI & Analytics ambitions and realizations are consistent with all organization policies (especially security, compliance, and digital sustainability policies).
Design & Develop BI & Analytics Value Chain	
Define BI & Analytics Architecture Principles and Ensure Consistency of Solutions	Define BI & Analytics architecture principles and ensure the consistency of solutions.
Design Infrastructure & Adapted Storage Spaces	Design the Technical Architecture for BI & Analytics purposes, including the structuration of relevant storage spaces fitting various requirements and uses (cold data, hot data, sensitive data, etc.).
Model & Build Data Structuration Depending on Related Uses (Raw, Warehouse, Business Views)	Model and implement the structuration of data to support the whole BI & Analytics value chain, from raw data to business views.
Define & Implement Business Rules	Define and implement rules related to business requirements (business calculation rules, business data validity checks, etc.).
Define & Implement Technical Rules	Define and implement technical rules. Typically, it is about basic technical format rules (e.g., homogeneity of data, float (...) formats).
Define & Implement Specific Data Protection Rules	Define and implement specific data protection rules, including retention rules, compliance automation, etc.
Design & Develop Data Pipelines	Design and develop flows from raw data to business views.
Design Semantic Layer	Design the semantic layer, so that data should be easily manipulated by business users; for example, for <i>ad hoc</i> /self-service reports.
Data Wrangling	Explore data, visualize available data for analytics purposes, build first sample machine learning models (for exploration purposes).
Design & Build Advanced Analytical Models	Define with business what event or problem must be modeled, for what purpose, and with which operational requirements, design and build advanced analytical models to meet the requirements, and evaluate the model regarding these requirements.

Business Function	Description
Design & Build Industrialized Stable Reports	Design and build stable recurrent reports.
Design & Build <i>Ad Hoc</i> Reports	Design and build <i>ad hoc</i> reports; for example, on a “self-service” mode – this activity is typically directly done by a business actor, final user.
Industrialize & Deploy Advanced Analytical Models	Operationalize a validated advanced analytical model (e.g., a classification model), integrate the model in the related business process (e.g., for optimization purposes), put in place a way to monitor and control the accuracy in real life.
Provide Necessary Data Sources for BI & Analytics Purposes	
Identify Relevant Data Sources	Develop practices and processes to identify quickly and precisely the relevant data sources for a given use case, and qualify these sources.
Identify & Manage Data Providers	Manage technical and business relationships with external data providers.
Design & Build Necessary Interfaces	Design and build necessary data interfaces to support the BI & Analytics value chain; e.g., feed raw data spaces from data sources, align operational systems with indicators calculated by BI & Analytics capability, etc.
Govern & Measure	
Define Governance Organization & RACI	Define governance organization, especially including the roles of business actors and CDO and IT teams in the build and run of the BI & Analytics capability.
Define & Enforce Process to Onboard New Source Data	Define the data onboarding process: what should be done when a new data source enters the BI & Analytics scope (e.g., qualification of the new source, qualification of the related data, add the new source in the metadata management repository, BI & Analytics usages specifications, etc.).
Define & Enforce Cost Optimization Principles	Define how to optimize costs of the BI & Analytics capability (optimization principles around organization, processes, technical platform implementation, and monitoring, etc.).
Define & Enforce Compliance Requirements	Define and enforce compliance requirements.
Measure Data Volumes	Be aware of the data volumes necessary for BI & Analytics purposes (useful to monitor and control the costs, to encourage digital sustainability, etc.).
Measure Data Use Frequencies	Be aware of which data is used and which data is poorly used; it is key information to understand the value of existing data in the BI & Analytics platforms.
Follow Up ROI of Usages	Measure the value of BI & Analytics usages through Return on Investment (ROI) definition and calculation.
Protect Data & Ensure Compliance	
Define & Manage Data Access	Define, manage, and control the way data can be accessed, how, and by whom.
Audit Platform Security & Compliance	Set up a recurring audit on BI & Analytics platforms, processes, security, and compliance.
Define Anonymization Requirements & Processes	Define the reasons why and how data should be anonymized/pseudonymized.
Detect & Notify a Data Breach	Put a process in place to detect and notify a potential data breach.

Business Function	Description
Support for Handling Data Portability Requests	Having an organization to support the handling of data portability requests.
Define & Apply a Data Quality Framework (Measures, Quality Management, etc.)	Apply a data quality framework for BI & Analytics purposes.
Leverage Metadata Management for BI & Analytics	Apply a metadata management capability to the BI & Analytics data value chain.
Operate Platforms	
Detect & Manage Incidents	Detect and manage incidents on BI & Analytics platforms/tools.
Detect & Manage Data Issues	Detect and manage data issues (data quality, data pipeline issues, etc.).
Monitor & Control Costs	Monitor costs, especially for cloud-based solutions.
Monitor & Control Data Quality	Monitor quality of data used for BI & Analytics use cases.
Monitor & Control Model Accuracy	Monitor the accuracy of deployed advanced analytical models.
Schedule Data Integration & Pipeline	Manage all scheduled tasks for BI & Analytics purposes (data pipeline runs, integration, etc.).
Grant Access	Handle access demands/grant access according to compliance and security policies.
Communicate & Train	
Train End Users in BI & Analytics & Tools	Organize training so that each business and data expert is comfortable in their BI & Analytics practice.
Leverage Technical Evangelism	Encourage capitalization and the return of experiences between data experts (data scientists, data engineers, data managers, etc.). Leverage reusable materials.
Animate BI & Analytics Users & Expert Communities	Animate user communities (on your BI & Analytics platforms tools, for example), data expert communities.
Communicate on Strategy, Business, & Technical Roadmaps	Communicate on BI & Analytics strategy, value, key ongoing/finished projects, and business and technical roadmaps; communicate on the impact of BI & Analytics capability for the organization.

4.3. BI & Analytics Reference Model – Detailed Application Functions

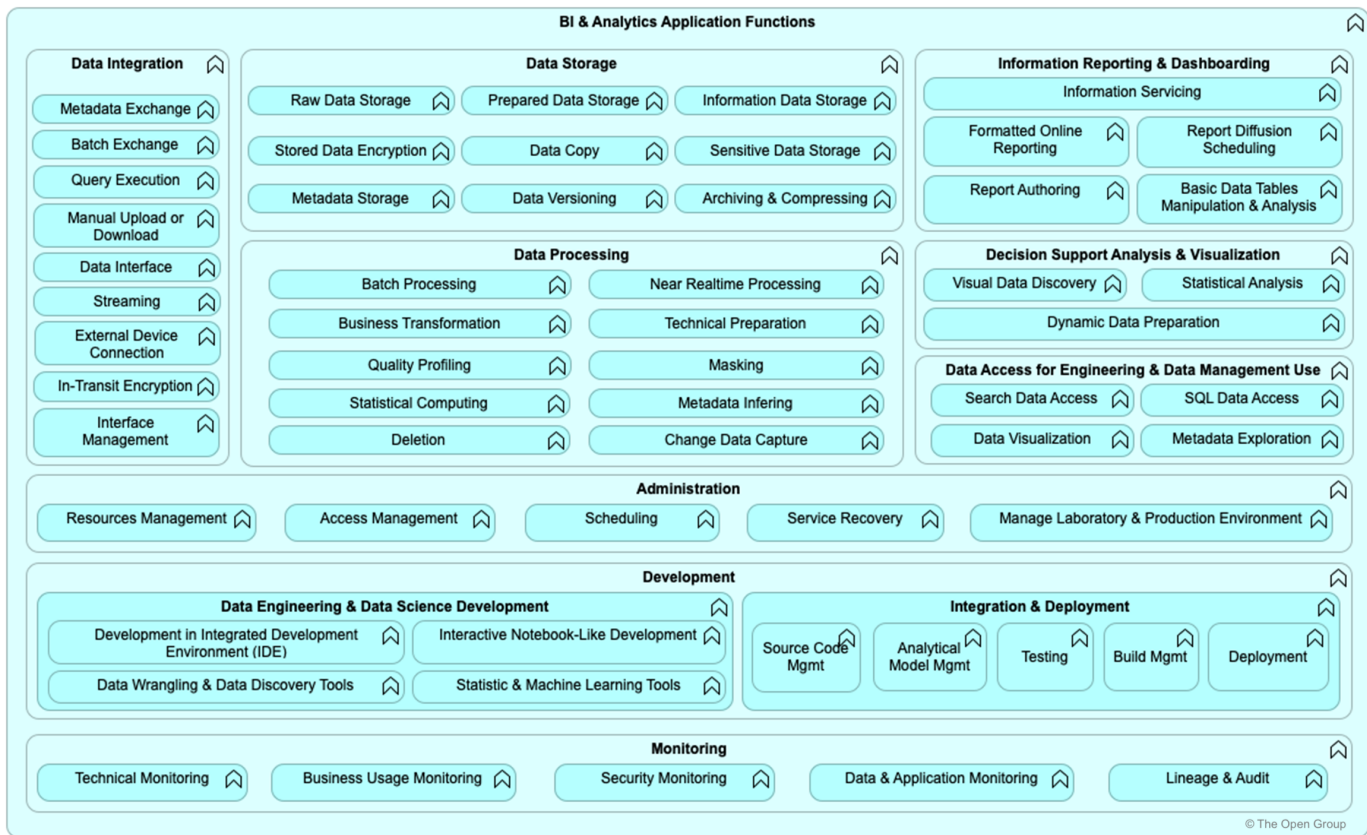


Figure 4-3: BI & Analytics Reference Model – Detailed Level

4.3.1. Description of the Application Functions

Table 4-3: Detailed Application Functions

Application Function	Title	Capabilities Applicable to the Full Scope of BI & Analytics
Administration Administration tasks required to support BI & Analytics capabilities such as access management, scheduling data processing, etc. Requirements listed here apply to the full scope.	Resources Management	Resources management of BI & Analytics
	Access Management	Identity and access management, mainly including authentication and authorization services for users and administrators
	Scheduling	Job scheduler administration
	Service Recovery	Recovery of the service after a disaster
	Manage Laboratory & Production Environment	Management of different environments: production, test, development

Application Function	Title	Capabilities Applicable to the Full Scope of BI & Analytics
Data Access for Engineering & Data Management Use Direct access to the data of BI & Analytics for advanced data stewards, data engineers, data scientists, data administrators, etc. Requirements listed here apply to the full scope.	Search Data Access	Ability to search for data through a search engine directly on BI & Analytics storage
	Data Visualization	Ability to visualize data directly on BI & Analytics storage
	SQL Data Access	Ability to perform Structured Query Language (SQL) queries directly on BI & Analytics storage
	Metadata Exploration	Ability to explore the metadata of BI & Analytics
Data Engineering & Data Science Development Capabilities applicable to the developer's workstation.	Development in Integrated Development Environment (IDE)	IDE for the data engineer or data scientist
	Data Wrangling & Data Discovery Tools	Data preparation tooling for the data engineer or data scientist
	Interactive Notebook-Like Development	Notebook environment (for data analytics in early exploration phase, algorithm comparison) for data scientists, data engineers, and graphical presentation of data
	Statistic & Machine Learning Tools	Statistic and machine learning tools for data scientists and data engineers
Integration & Deployment Continuous delivery chain, including versioning, deployment, packaging, etc.	Source Code Management	Versioning of development code (trunk) and release (branch) for developers or data scientists
	Analytical Model Management	Functionalities to manage life-advanced, analytical models (e.g., machine learning models) in their end-to-end lifecycle
	Testing	Execution of tests on the new code release (non-regression)
	Build Management	Prepare the package to be deployed: run the test and generate binaries
	Deployment	Deploy a package – new software release – on a given environment (development, integration, pre-production, production)

Application Function	Title	Capabilities Applicable to the Full Scope of BI & Analytics
Data Integration Services that allow integration and BI & Analytics in the full landscape including batch, streaming, interfaces, etc. Requirements listed here apply to the full scope.	Metadata Exchange	Metadata ingestion and distribution in parallel from data
	Batch Exchange	Scheduled asynchronous distribution of files
	Query Execution	Synchronous Java® Database Connectivity (JDBC) or Open Database Connectivity (ODBC) database connection
	Manual Upload or Download	Integration of a file provided by an end user, or by downloading a set of data
	Data Interface	Web service integration, APIs, etc.
	Streaming	Streaming through a messaging system that transports and secures data flow back pressure
	External Device Connection	Integration with an external device connection
	In-Transit Encryption	Protection of data transport through the encryption of the data in motion (transit) on the network
	Interface Management	Capabilities applicable to full “Interface Management” scope to provide a central point to monitor and configure interfaces in a self-service mode for entities
Data Processing Services that allow data transformation including basic preparations or more advanced feature calculations leveraging statistics or data science. Requirements listed here apply to the full scope.	Batch Processing	Daily, weekly, monthly, and yearly data transfers performed by batch
	Business Transformation	Business processing (filtering, aggregation, joining, and de-duplication)
	Quality Profiling	Profiling of data quality through statistical techniques
	Statistical Computing	Data science and statistical processing (classification, clustering of data using statistics, data science modeling, including machine learning)
	Deletion	Deletion of data from BI & Analytics (e.g., for GDPR)
	Near Realtime Processing	Low-latency data processing – less than 24 hours a day (includes micro-batch and near-real time, with minute or second latency)
	Technical Preparation	Business-agnostic technical processing common to all use cases (i.e., type format, cleansing, integration)
	Masking	Anonymizing or pseudonymizing BI & Analytics data
	Metadata Infering	Process of deducing properties of data – such as type, classification, metadata, etc. – through the analysis of the data itself
	Change Data Capture	Ingestion or distribution of change data capture flows

Application Function	Title	Capabilities Applicable to the Full Scope of BI & Analytics
Data Storage Services that allow persistent storage of structured or unstructured data. Requirements listed here apply to the full scope.	Raw Data Storage	Storage of source data without any change
	Stored Data Encryption	Protection of data within a node through encryption of the data at rest
	Metadata Storage	Storage of the metadata required by BI & Analytics (schema)
	Prepared Data Storage	Storage of data that has been processed applies standard technical rules that are business-agnostic: type formats, management of null value, generation of unique keys, etc.
	Data Copy	Function that allows copying of datasets between different environments
	Data Versioning	Functions available to facilitate the versioning of data collections
	Information Data Storage	Storage of data that has been prepared for a specific business use case: insights, features calculated by data science, aggregations, includes the capability to execute low-latency queries for streaming jobs
	Sensitive Data Storage	Storage of sensitive data; e.g., personal, financial, strategy, business, etc.
	Archiving & Compressing	Archiving and compression functions to manage the data lifecycle and optimize storage resources
Decision Support Analysis and Visualization Support data discovery and visualization using data stored in BI & Analytics.	Visual Data Discovery	Dynamic drill down to navigate the data and find the right chart, graphic, etc.
	Statistical Analysis	Apply dynamic statistical “off-the-shelf” models to information available online
	Dynamic Data Preparation	Data dynamic preparation, with no persistence, to allow statistical processing
Information Reporting & Dashboarding Support reporting and dashboarding activities using data stored in BI & Analytics.	Information Servicing	Storage of information data in a format adapted to queries (cube, in memory, view, etc.)
	Formatted Online Reporting	Ready-to-use reports and charts that can be refreshed interactively to use data with no specific data management skills
	Report Authoring	Build reports and refresh them with data, with the ability for a business user – with no SQL skill, and a drag-and-drop User Interface (UI) – to create a report from available information and share it with other users
	Report Diffusion Scheduling	Ready-to-use static reports, charts generated periodically, usually paginated and in a printable format
	Basic Data Tables Manipulation & Analysis	Functionalities to facilitate the most basic and frequent manipulation, treatment, and analysis of data tables

Application Function	Title	Capabilities Applicable to the Full Scope of BI & Analytics
Monitoring Monitoring services for infrastructure, network, and application to guarantee the agreed service level.	Technical Monitoring	Monitoring of the network layer and infrastructure layer
	Business Usage Monitoring	Monitoring of the business services of the platform (supervise data science model drifting, etc.)
	Security Monitoring	Monitoring of the security layer
	Data & Application Monitoring	Monitoring of the data (e.g., quality) and application layer (e.g., availability of services)
	Lineage & Audit	Capture of the data lifecycle, including the data origin and where it moves over time

4.4. BI & Analytics Reference Model – Requirements

To complement this document, a spreadsheet file with an example requirements list for a BI & Analytics capability is provided.

The architecture requirements listed in this file give some details about the application functions that are described in the reference architecture.

It should probably not be read as a whole, but can be a reference to find details on some functions.

Architecture Requirements

The Enterprise Architecture systemic approach is applied here. The whole scope of work is a system composed of subsystems. Each function constitutes a subsystem. Thereby, the requirement wording “the system” refers to the function of the requirement.

They are provided in the spreadsheet file by the functions listed and here, in this chapter, they are provided in alphabetical order.

The architecture requirements represent formal statements of need, as expressed by stakeholders, which must be met by the architecture.

Architecture requirements capture what is architecturally significant or, in other words, they should capture what has an impact on the architecture decisions. They should not be as detailed as specification or test requirements, but can be an input to writing them.

By formal statements we mean:

- Explicit and understandable need
- Not solution-oriented, no tooling name
- Atomic
- Unique
- Verifiable

The architecture requirements are synthetic but detailed enough to allow:

- Clear understanding of the capabilities needed and delivered
- Roadmap prioritization
- Securing of the implementation of the BI & Analytics platform
 - Architecture decisions
 - Precise understanding of the capabilities provided by IT suppliers

The formalization of requirements uses the following structure: “The system should [verb][object][stakeholder]”.

A: Glossary

ADM

Architecture Development Method

API

Application Programming Interface

Simplify software and integration architecture developments by enabling applications to exchange data and functionalities easily through structured interfaces.

ASF

Apache Software Foundation

BI

Business Intelligence

CDO

Chief Data Officer

Key role in organization responsible for measuring and increasing value of data in the organization through data management, data architecture, data governance, analytics use cases, etc.

CRM

Customer Relationship Management

Data Mart

A structure/access pattern specific to data warehouse environments, used to retrieve client-facing data. A subset of the data warehouse and is usually oriented to a specific business line or team. Whereas data warehouses have an enterprise-wide depth, the information in data marts pertains to a single department. Doing so, it decouples the different business BI & Analytics solutions and allows step-by-step upgrade.

Enterprise Data Warehouse

A subject-oriented, integrated, time-variant, non-volatile collection of data in support of management's decision-making process [[Inmon, 1995](#)].

ERP

Enterprise Resource Planning

ETL

Extract, Transform, Load

Usually names the data integration process that combines data from one or multiple sources into a single, consistent data store. Also refers to a certain type of solutions supporting this process.

GDPR

General Data Protection Regulation

Regulation that harmonizes data privacy laws across Europe.

IDE

Integrated Development Environment

Software suite that consolidates basic tools required to write and test software.

Infocenter

The first generation of BI & Analytics capabilities set up in the 1970s and 1980s. A raw copy of the mainframe operational systems data was made available for analytics by business users. It was then replaced by data warehouses, but they are still used and in place in some organizations.

JDBC

Java Database Connectivity

KPI

Key Performance Indicator

NPS

Net Promoter Score

ODBC

Open Database Connectivity

ODS

Operational Data Store

A database designed to integrate data from multiple sources for additional operations on the data. Unlike a master data store, the data is not passed back to operational systems. It may be passed for further operations and to the data warehouse for reporting. Because the data originates from multiple sources, the integration often involves cleaning, resolving redundancy, and checking against business rules for integrity. It is usually designed to contain low-level or atomic (indivisible) data (such as transactions and prices) with limited history that is captured in “real time” or “near-real time” as opposed to the much greater volumes of data stored in the data warehouse generally on a less-frequent basis. The general purpose is to integrate data from disparate source systems in a single structure, using data integration technologies like data virtualization, data federation, or ETL. This will allow operational access to the data for operational reporting, master data, or reference data management. It is not a replacement or substitute for a data warehouse, but can be a source.

P&L

Profit and Loss

RACI

Responsible, Accountable, Consulted, Informed

RDBMS

Relational Database Management System

ROI

Return on Investment

SQL

Structured Query Language

A standardized programming language that is used to manage relational databases and perform various operations on the data in them.

Staging Area or Landing Zone

An intermediate storage area used for data processing during the ETL process. The data staging area sits between the data source(s) and the data target(s), which are often data warehouses, data marts, data lakes, or other data repositories. Data staging areas are often transient in nature, with their contents being erased prior to running an ETL process or immediately following successful completion of an ETL process.

TCO

Total Cost of Ownership

UI

User Interface

Index

A

application function, [2](#)
application functions, [6](#)

B

business function, [2](#)

D

data marts, [14](#)
data warehouse, [7](#)

E

enterprise data warehouse, [8](#)

G

GDPR, [3](#)
goal, [1](#)

I

IDE, [4](#)
Infocenter, [7](#)

J

JDBC, [27](#)

L

legacy architecture, [7](#)

O

ODBC, [27](#)
ODS, [13](#)

P

principles, [7](#)

S

stakeholder, [2](#)

T

transactional systems, [8](#)
Transition Architecture, [7](#)